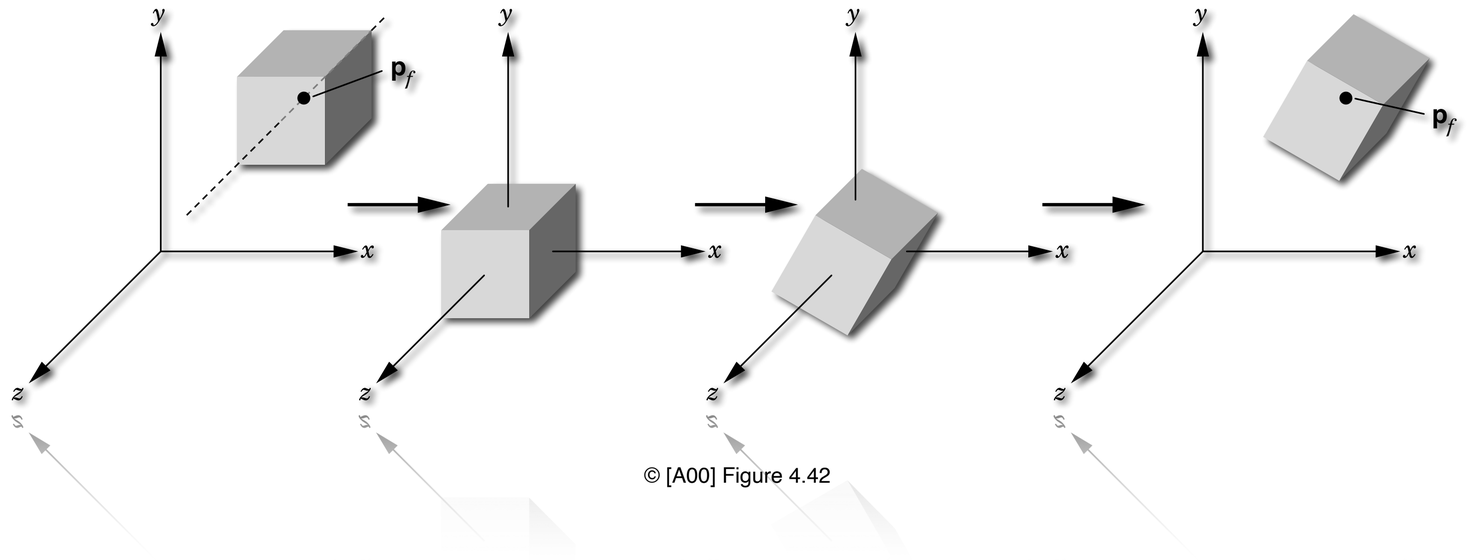
Compositions of Transformations



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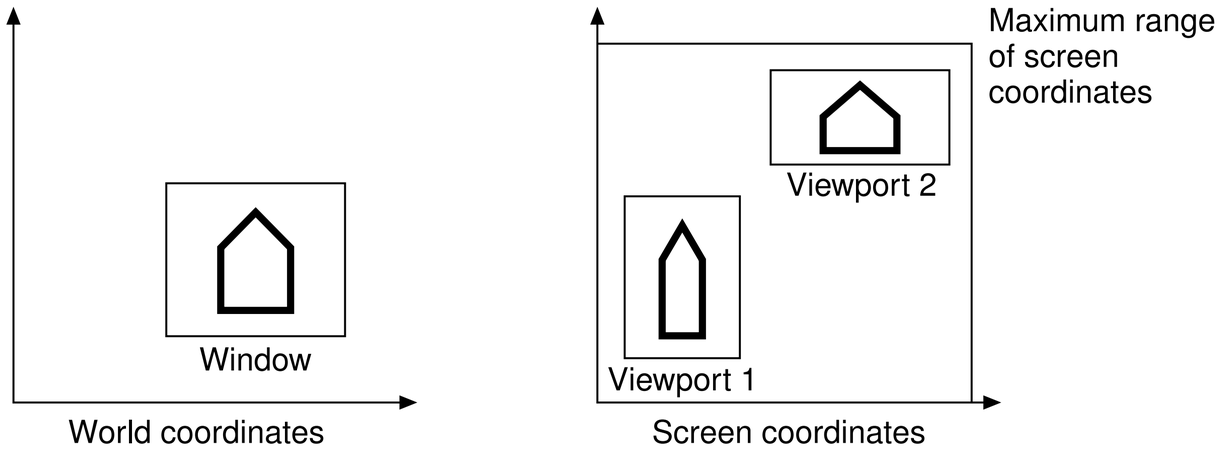
Composition of Transforms

- Multiple fundamental transformations T_i can be composed into one single transformation by matrix multiplication:

$$M = T_\mathrm{k} \cdot T_\mathrm{k\text{-}1} \ldots T_2 \cdot T_1$$

  - note the right-to-left order, $\boldsymbol{T_i}$ is applied before $\boldsymbol{T_{i+1}}$ on the object
  - gain efficiency by combining transformations applied to many objects or elements
  - order of transformations is important as matrix multiplication is **NOT** commutative in general
    - only very few specific transformations (e.g. **uniform** scaling) can partially be reordered

2D Example: Window-to-Viewport



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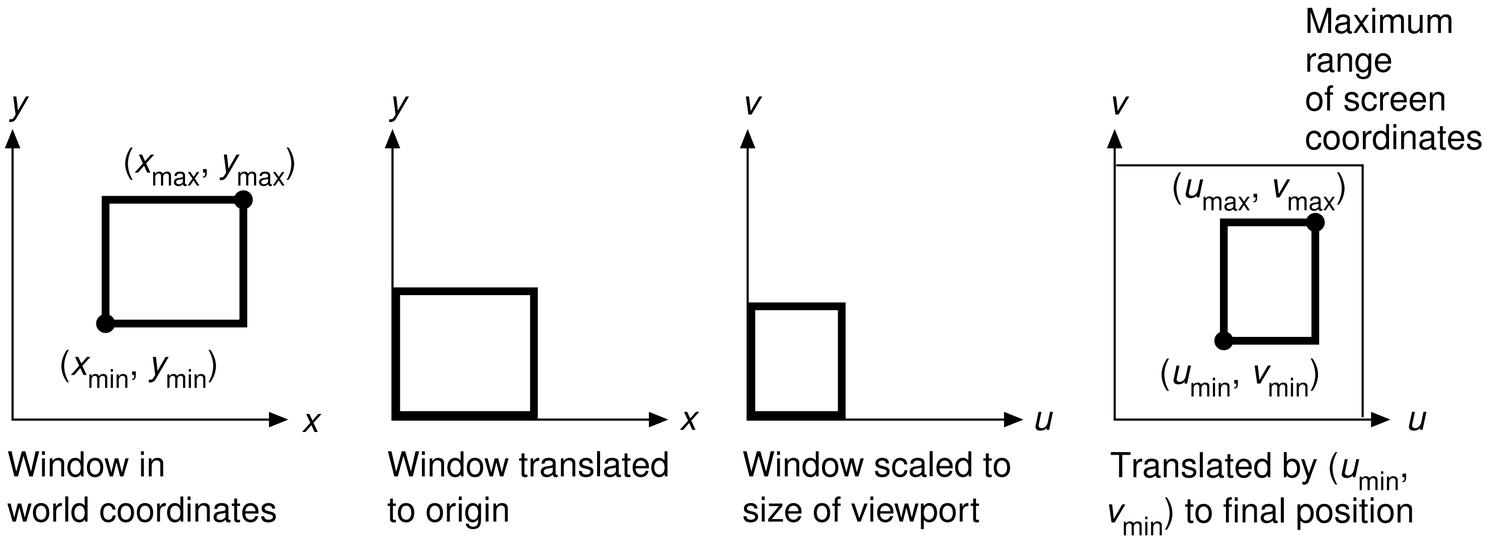
- Map the 3D scene's window **coordinates** into device or **screen coordinates**
  - specified by world-coordinate window and corresponding rectangular viewport region on screen
    - applied to all output primitives in world window coordinates
  - viewing window and viewport may have different **aspect ratios**

Window-to-Viewport Mapping



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- Window-to-viewport mapping matrix M_{WV} is composed of:
  1. Translation of world coordinate window to origin
  2. Scaling to size of viewport rectangle
  3. Translation of viewport to final position on screen

$$M_{wv} = T(u_{min}, v_{min}) \cdot S\left(\frac{u_{max}-u_{min}}{x_{max}-x_{min}}, \frac{v_{max}-v_{min}}{y_{max}-y_{min}}\right) \cdot T(-x_{min}, -y_{min})$$

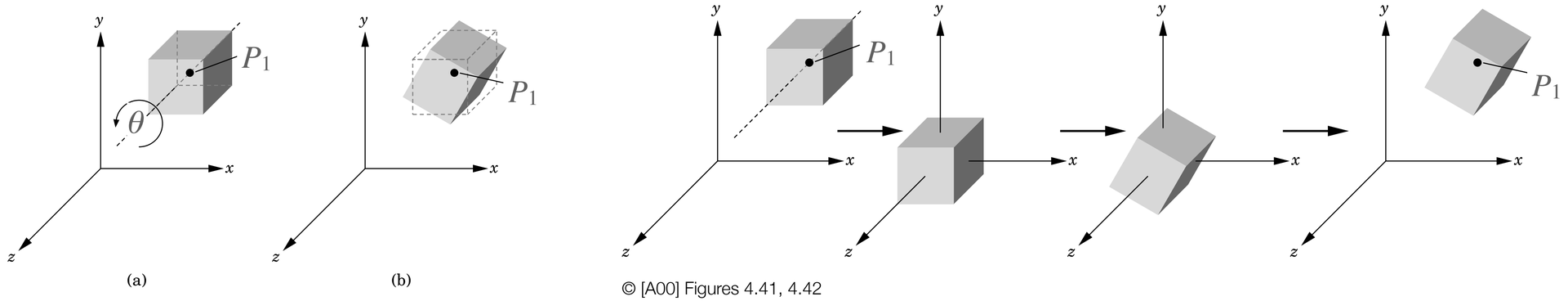
$$= \begin{bmatrix} 1 & 0 & u_{min} \\ 0 & 1 & v_{min} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \frac{u_{max}-u_{min}}{x_{max}-x_{min}} & 0 & 0 \\ 0 & \frac{v_{max}-v_{min}}{y_{max}-y_{min}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_{min} \\ 0 & 1 & -y_{min} \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{u_{max}-u_{min}}{x_{max}-x_{min}} & 0 & -x_{min} \cdot \frac{u_{max}-u_{min}}{x_{max}-x_{min}} + u_{min} \\ 0 & \frac{v_{max}-v_{min}}{y_{max}-y_{min}} & -y_{min} \cdot \frac{v_{max}-v_{min}}{y_{max}-y_{min}} + v_{min} \\ 0 & 0 & 1 \end{bmatrix}$$

- Transformation of points $P=(x,y,1)$ by M_{WV} yields expected result:

$$P' = \left((x - x_{min}) \cdot \frac{u_{max}-u_{min}}{x_{max}-x_{min}} + u_{min}, (y - y_{min}) \cdot \frac{v_{max}-v_{min}}{y_{max}-y_{min}} + v_{min}, 1\right)$$

Composition Example 1



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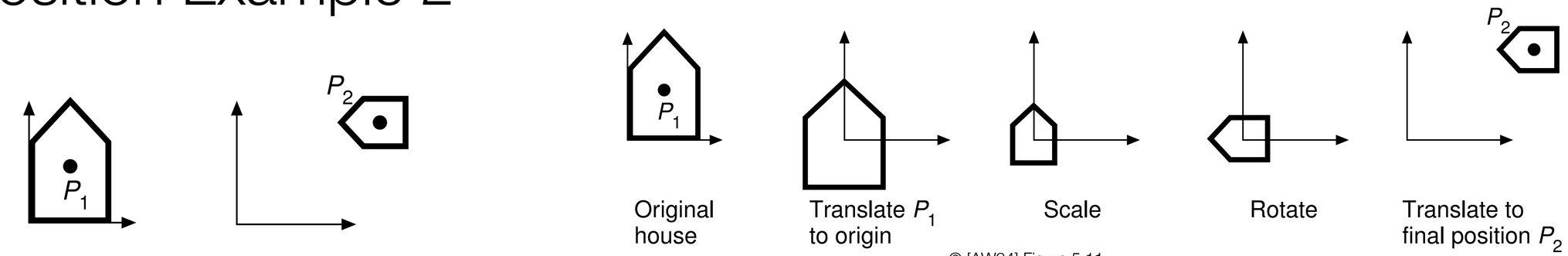
- Rotation about arbitrary P_1 can be composed from a few consecutive fundamental transformations
  1. Translation of P_1 to the origin
  2. Rotation e.g. about z-axis (at the origin)
  3. Translate P_1 back to its initial location
- *2D Example*: Rotation origin is (x_1, y_1) and rotation angle θ: $T(x_1, y_1)\cdot R(\theta)\cdot T(-x_1,-y_1)$

$$M = \begin{bmatrix} 1 & 0 & x_1 \\ 0 & 1 & y_1 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_1 \\ 0 & 1 & -y_1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & x_1(1-\cos\theta)+y_1\sin\theta \\ \sin\theta & \cos\theta & y_1(1-\cos\theta)-x_1\sin\theta \\ 0 & 0 & 1 \end{bmatrix}$$

Composition Example 2



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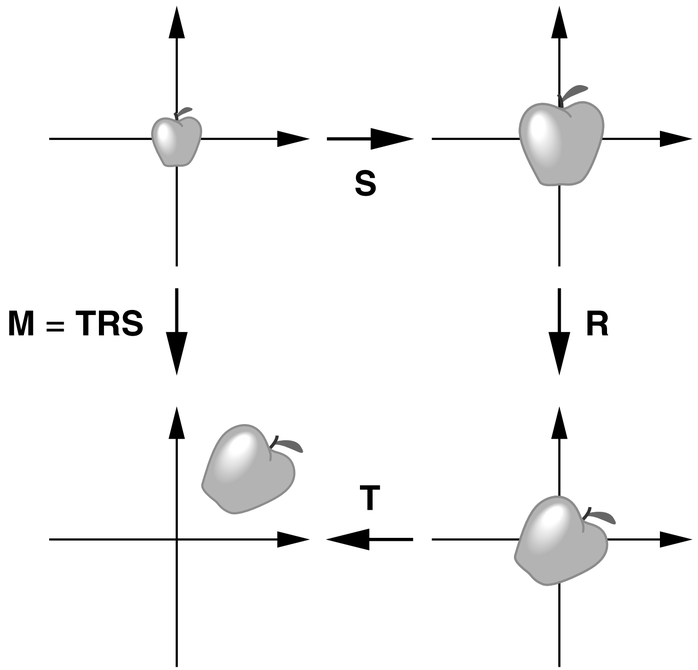
- Scaling **and** rotation about arbitrary anchor point
  - anchor point should not undergo any side-effect translation
  1. Translation of P_1 to the origin
  2. Scaling by s_x and s_y
  3. Rotation 90° counter clock-wise
  4. Translate back to (new) location P_2
- *2D Example*: Scaling origin is center point (x_1, y_1): $T(x_2, y_2) \cdot R(90°) \cdot S(s_x,s_y) \cdot T(-x_1,-y_1)$

$$M = \begin{bmatrix} 1 & 0 & x_2 \\ 0 & 1 & y_2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_1 \\ 0 & 1 & -y_1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -s_x & x_1 + s_x y_1 \\ s_y & 0 & y_1 - s_y x_1 \\ 0 & 0 & 1 \end{bmatrix}$$

SRT Standard

- Typical combination of transformations includes selecting the scale of the object, its orientation in space and position of the object at the desired location

- SRT: **scale – rotate – translate**
  - in matrix order (right to left!): $\boldsymbol{T \bullet R \bullet S}$
  - but possibly first set scale/rotation anchor point (x_1, y_1)



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Implementation

- Any translation, scaling, and rotation transformations can be combined together in a **composite matrix**
  - **upper-left 3x3** submatrix R is

    aggregate rotation and scaling
  - **upper-right 3x1** column vector T

    represents **aggregate translation**

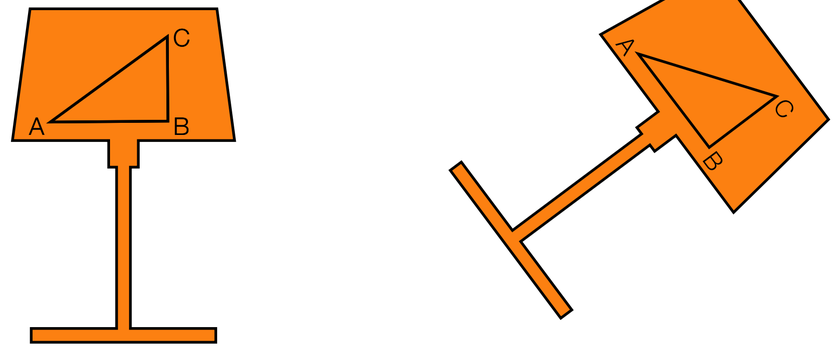
$$M = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = T \cdot R \cdot S$$

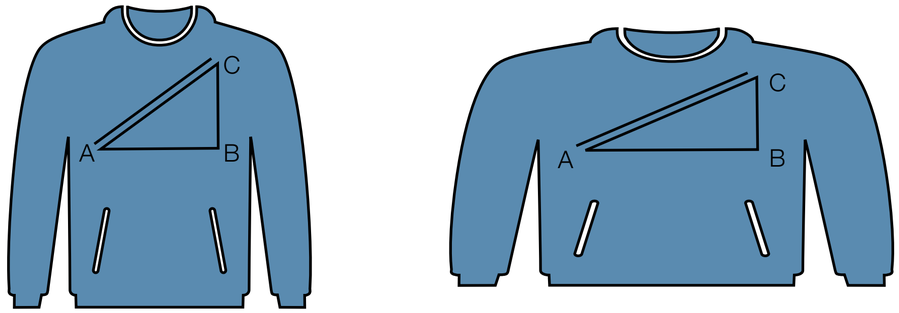
- Apply rotation/scale and translation separately for efficiency
  - 9 multiplications and 9 additions (instead of 16 and 12)

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = R \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} + T.$$

Transformation Properties

- **Rigid-body** transformations
  - preserving lengths and angles, no object distortion
- **Affine** transformations
  - preserving parallelism
  - non-uniform scaling is affine but not a rigid-body transform
- Scaling and rotation are dependent on the origin
  - translation is not
  - scaling (and rotation) is an operation on vectors
    - or on position vectors
  - translation is an operation on points





Inverse

- The transformation, scaling, and rotation matrices all have inverses
  - not true for all general matrices
- The inverses are formed by:

  negation of displacements $d_x, d_y, d_z \rightarrow -d_x, -d_y, -d_z$

  reciprocal of scale values $s_x, s_y, s_z \rightarrow 1/s_x, 1/s_y, 1/s_z$

  negation of rotation angle $\theta \rightarrow -\theta$

- Inversion **of any orthogonal matrix** B is equivalent to **transposition** $B^{-1}=B^T$
  - works well for the special orthogonal rotation matrices
  - consistent with negation of rotation angles

Inverse Composition

- Inversion of a sequence of transformations is done by:
  inverting the **sequence order** and
  inverting each **transformation** individually

$$M = M_4 \cdot M_3 \cdot M_2 \cdot M_1$$

$$M^{-1} = M_1^{-1} \cdot M_2^{-1} \cdot M_3^{-1} \cdot M_4^{-1}$$

Exercise:Transformation of Object Normals

- Geometric objects defined by points are transformed by transforming their points directly:

$$P' = M \cdot P$$

- **Transformation of a plane normal (or plane coefficients) is different**
  - plane **equation** $\boldsymbol{Ax + By + Cz + D = 0}$ defined by generalized plane parameters $N = (A, B, C, D)$ and for any point $P = (x, y, z, 1)$ on the plane
  - for any point $P = (x, y, z, 1)$ on the plane it is **equivalent to** $\boldsymbol{N^{\mathrm{T}} \cdot P = 0}$, using the dot-product of N and P

- Normal transformed by Q as: $N' = Q \cdot N$
  - transformation Q of normals is not the same as M
  - it must maintain that $\boldsymbol{N'^{\mathrm{T}} \cdot P' = 0}$ for all transformed points $P' = M \cdot P$

Exercise:Transforming Object Normals

- The plane equation

$$N'^{\mathrm{T}} \cdot P' = (Q \cdot N)^{\mathrm{T}} \cdot M \cdot P = N^{\mathrm{T}} \cdot Q^{\mathrm{T}} \cdot M \cdot P = 0$$

  holds iff $Q^{\mathrm{T}} \cdot M$ **is a multiple of the identity matrix**

  - Using $(Q \cdot N)^{\mathrm{T}} = N^{\mathrm{T}} \cdot Q^{\mathrm{T}}$, and since $N^{\mathrm{T}} \cdot P = 0$

- Thus from $Q^{\mathrm{T}} \cdot M = I$ we get that $Q = (M^{-1})^{\mathrm{T}}$
  - note that M^{-1} may not exist in general due to zero determinant (e.g. if including a projection)
  - however, if M consists only of fundamental transforms (such as e.g. translate, scale, rotate, shear, reflect) then M^{-1} exists and can be formed knowing the composition of M
- **Transform points by M, and transform normals by Q**
  - transform vectors correctly by transforming its endpoints

Transformation Task

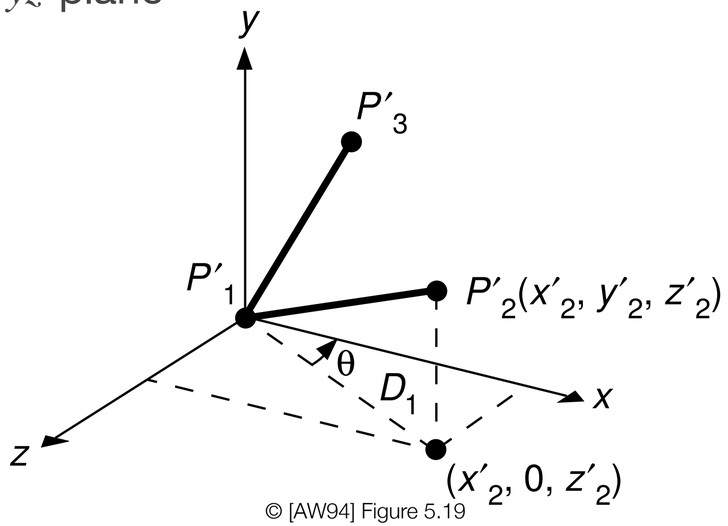
(a) Initial position

(b) Final position

- Transform P_1 to origin, P_2 onto z-axis and P_3 into yz-plane using rigid body transforms
- **First approach**: compose from four fundamental transforms
  1. translate P_1 to origin
  2. rotate around the y-axis to move P_2 into the yz-plane
  3. rotate around the x-axis to align P_2 with the z-axis
  4. rotate around the z-axis to move P_3 into the yz-plane

Transformation Task 1a

- Translation (1): $T(-x_1, -y_1, -z_1)$ is simply given by the negative coordinates of P_1
- Rotation (2): $R_y(\theta\text{-}90)$ is defined by the angle θ between the projection D_1 of P_1'P_2' in the xz-plane and the x-axis
  - rotate about y-axis to map P_2 into the yz-plane



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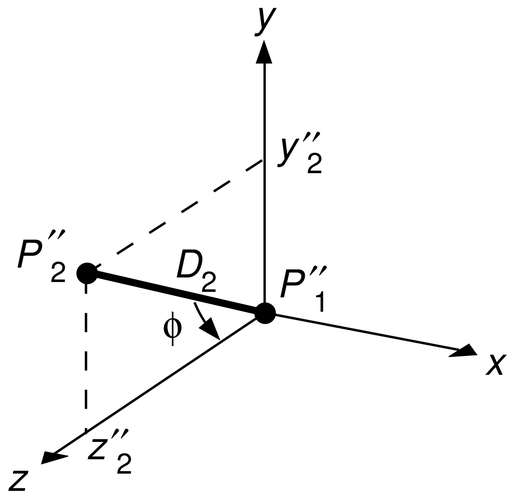
Transformation Task 1a

- Rotation (3): $R_x(\phi)$ is defined by the angle ϕ of $D_2 = P_1'' P_2''$ in the the yz-plane and the z-axis
  - rotate about x-axis to align P_2 with z-axis



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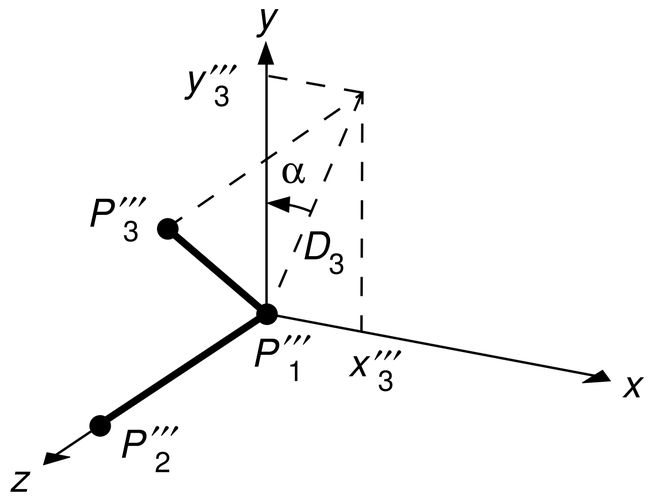
Transformation Task 1a

- Rotation (4): $R_z(\alpha)$ is defined by the angle α between the projection D_3 of $P_1'''P_3'''$ in the xy-plane and the y-axis
  - rotate about z-axis to map P_3 into yz-plane



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- **Final composite:** $M = R_z(\alpha) \cdot R_x(\phi) \cdot R_y(\theta - 90) \cdot T(-x_1,-y_1,-z_1)$

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